GIOVANNI JONATHAN R. CRUZ

COMPUTER SCIENCE STUDENT | SOFTWARE ENGINEERING

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My Portfolio LinkedIn Github

EDUCATION

Bachelor of Science in Computer Science Major in Software Technology **2022 - Present**

De La Salle University | Manila

TECHNICAL SKILLS

- **Languages**: C, C#, C++, Python, JavaScript, SQL
- **Web & Frameworks**: Django, React, Node.js, TypeScript
- **Tools**: Git, GitHub, Figma, Framer, ShapesXR

PROJECTS

DrumXRoll — XR Drum Improvisation & Learning System **2024 – Present**

Thesis Project

- Designed and developed an XR-based drum learning system focused on improvisation, timing, and embodied musical flow
- Implemented a piano-roll–style visualization mapped to physical drum inputs to support real-time feedback and coordination
- Addressed XR-specific challenges such as latency sensitivity, feedback clarity, and immersion breakdown
- Explored how system responsiveness affects perceived “feel” versus accuracy in musical performance
- Built as a research-driven project combining software engineering, interaction design, and experiential evaluation

Distributed Database System — Advanced Database Project **Aug 2024 – Dec 2024**

Course Project

- Designed and implemented a distributed database system using Django with multi-master replication
- Implemented synchronization mechanisms and transaction isolation handling to support high concurrency
- Explored consistency trade-offs and concurrency challenges in distributed environments
- Gained hands-on experience with replication, conflict handling, and system reliability

Emberlands — Emotional Mapping & Reflective Visualization **Nov 2025 - Dec 2025**

Personal Project

- Designed a system for mapping emotional states into interactive visual landscapes
- Focused on translating abstract emotional input into structured, explorable representations
- Prototyped interaction flows emphasizing reflection and interpretation
- Explored design trade-offs between emotional expressiveness and system structure
- Built as an exploratory project at the intersection of HCI, creative coding, and system design

LEADERSHIP & EXPERIENCE

Industry Partnerships Officer — Google Developers Student Club **Oct 2023 – Dec 2024**

- Coordinated partnerships with external organizations through proposal writing and professional correspondence
- Scheduled, documented, and followed up on meetings to improve internal coordination and communication
- Strengthened organizational and professional communication skills in a technical student organization